



UNIVERSITY OF
BIRMINGHAM

Superconductivity

Lecture 18 – “Attractive”

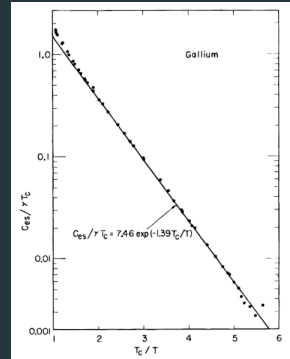
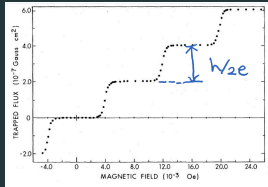
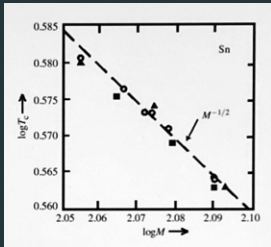
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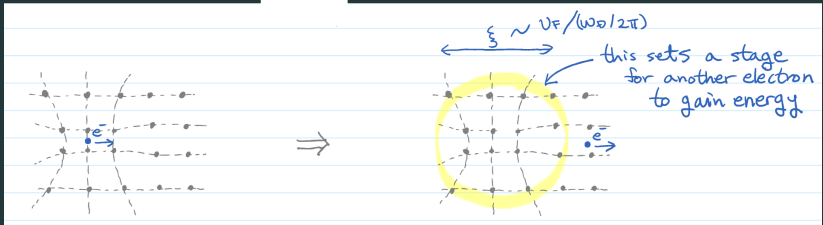
School of Physics & Astronomy

Lattice, Pairing and Gap



- Isotope effect $\Rightarrow$ lattice (phonon); θ_D or ω_D
- Flux quantum $\Rightarrow h/2e$
- Activated behaviour $\Rightarrow$ the gap $\Delta(T)$

Phonon-mediated electron pairing



- Fermi $T_F \sim 50 \times 10^3$ K; Debye $\theta_D \sim 300$ K.
- Fermi $v_F \sim 10 \times 10^3$ m/s; Debye $\omega_D \sim 5 \times 10^{13}$ rad/s
- Length scale: $v_F T = v_F (2\pi / \omega_D) \sim 0.1 \times 10^{-6}$ m $\sim$ coherence ξ
- Lattice relaxation time $\tau \sim 10^{-13}$ s (much slower than the electrons)

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Metal: A Fermi sea

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"Why metal $\rightarrow$ superconductor?"

What is metal?

- An ideal gas of fermions (electrons)
 $\Rightarrow$ A Fermi sphere

However,

- Electrons interact among each other (not a gas, but more a liquid)
- Electrons experience a lattice (periodic ionic potential)

Nevertheless,

- Electrons **screen** from each other
- Fermi sphere $\rightarrow$ **Fermi surface**

Metal vs. superconductor (2/2)

Let us stick to **metal = Fermi sphere** or sea as an approximation.

"Why metal $\rightarrow$ superconductor?" becomes

"Why Fermi sea $\rightarrow$ a collection of pairs?"

Want to show that

- E (paired electrons) $< E$ (Fermi sphere).

Note that

- At least, need a net attractive interaction to form a bound pair;
- Even with the attraction, forming a bound state is not obvious (e.g., kinetic energy).

The electronic screening: real space

Bare Coulomb interaction:

$$V(\mathbf{r}) = \frac{e^2}{4\pi\epsilon_0|\mathbf{r}|} \quad (1)$$

where $\mathbf{r}$ is a vector connecting two electrons. In a medium, this reduces by screening to

$$V_{\text{TF}}(\mathbf{r}) = \frac{e^2}{4\pi\epsilon_0|\mathbf{r}|} e^{-|\mathbf{r}|/r_{\text{TF}}}, \quad (2)$$

where the Thomas-Fermi length, r_{TF} , is order of atomic spacing.

The electronic screening: q space

$$V(\mathbf{q}) = \frac{1}{\Omega} \int V(\mathbf{r}) e^{i\mathbf{q} \cdot \mathbf{r}} d\mathbf{r} \quad (3)$$

$$= \frac{e^2}{4\epsilon_0 q^2}. \quad (4)$$

$$V(\mathbf{q}, \omega) = \frac{e^2}{4\epsilon_r(\mathbf{q}, \omega) \epsilon_0 q^2} \quad (5)$$

In the Thomas-Fermi approximation,

$$\epsilon_r = 1 + 1/q^2 r_{\text{TF}}^2 \quad (6)$$

The Coulomb repulsion is screened, thus reduced.

Free electron model

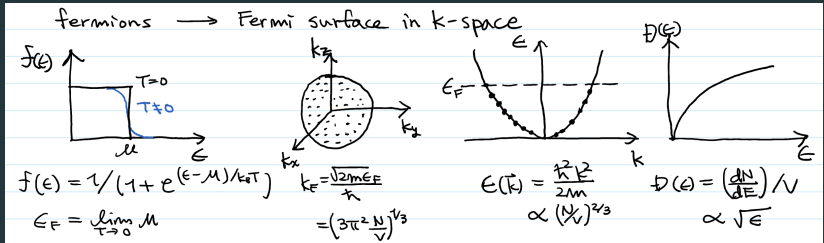


Figure 1: A summary for an ideal gas of fermions.

Nearly free electron model

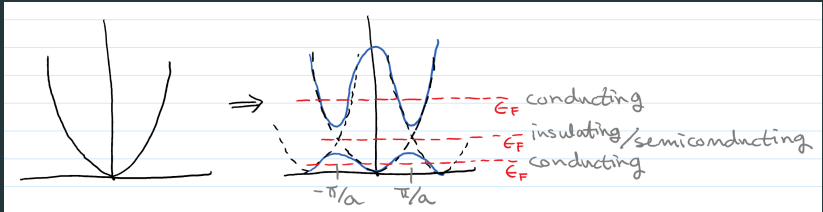


Figure 2: From a free electron model (left) to nearly free electron model with a periodic lattice (right) which allows distinction of transport properties of matter.

Two electrons interacting

Cooper's approach

How to show $E(\text{paired electrons}) < E(\text{Fermi sphere})$?

Cooper's idea: work out for **a single pair** of electrons to form a bound state in the background of Fermi sphere. To do so, **assume** that

- attractive 'net' interaction between the two electrons;
- the two electrons interact with the Fermi sphere through the exclusion principle only (no Coulomb).

Result: The Fermi sphere is unstable against forming a bound pair for **arbitrarily weak attractive interaction**.

Not at all trivial given that the electrons have kinetic energy ($\simeq \epsilon_F$)

Two attracted electrons (1/3)

The Hamiltonian:

$$\mathcal{H} = \frac{1}{2m_e} (\mathbf{p}_1^2 + \mathbf{p}_2^2) + V(\mathbf{r}_1 - \mathbf{r}_2), \quad (7)$$

The total wavefunction (orbital & spin):

$$\Psi(\mathbf{r}_1, \sigma_1, \mathbf{r}_2, \sigma_2) = \psi(\mathbf{r}_1, \mathbf{r}_2) \phi(\sigma_1, \sigma_2). \quad (8)$$

One may write

$$\psi(\mathbf{r}_1, \mathbf{r}_2) = \sum_{\mathbf{k}} g_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}_1} e^{-i\mathbf{k} \cdot \mathbf{r}_2}. \quad (9)$$

To respect the antisymmetry of Ψ , one may take $\cos$ term with spin-singlet, as

$$\psi(\mathbf{r}_1 - \mathbf{r}_2) = \left[\sum_{\mathbf{k}} g_{\mathbf{k}} \cos \mathbf{k} \cdot (\mathbf{r}_1 - \mathbf{r}_2) \right] \phi_{\text{singlet}}, \quad (10)$$

rather than taking $\sin$ term with a spin triplet.

Two attracted electrons (2/3)

By inserting Ψ and $\mathcal{H}$ to the Schrodinger equation, one obtains,
Evaluating the above will lead to

$$(E - 2\epsilon_k)g_k = \sum_{k' > k_F} V_{kk'} g_{k'}, \quad (11)$$

where

$$V_{kk'} = \frac{1}{\Omega} \int V(\mathbf{r}) e^{i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{r}} d\mathbf{r}. \quad (12)$$

Cooper assumed $V_{kk'} = -V$ for $E < E + \hbar\omega_D$ (0 for higher energy).
Then

$$g_k = V \frac{\sum g_{k'}}{2\epsilon_k - E} \quad (13)$$

which becomes

Two attracted electrons (3/3)

$$\frac{1}{V} = \sum_{k > k_F} (2\epsilon_k - E)^{-1} \quad (14)$$

Replacing the summation by an integration gives

$$\begin{aligned} \frac{1}{V} &= \int_{E_F}^{E_F + \hbar\omega_D} \frac{D(\epsilon)d\epsilon}{2\epsilon_k - E} \\ &= \frac{1}{2}D(E_F) \ln \frac{2E_F - E + 2\hbar\omega_D}{2E_F - E} \end{aligned}$$

For $D(E_F)V \ll 1$,

$$E \simeq 2E_F - 2\hbar\omega_D e^{-2/D(E_F)V} \quad (15)$$

A bound state exists since $E < 2E_F$

Electron-electron interaction via exchanging phonon

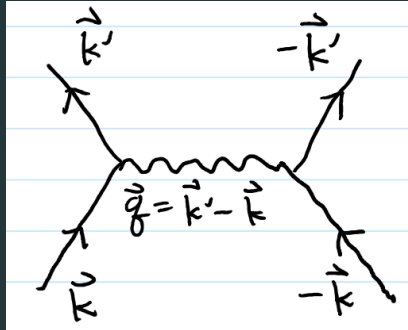


Figure 3: A diagram representing the interaction of two electrons via exchange of phonons.

Pairing via lattice with help of the Fermi sea

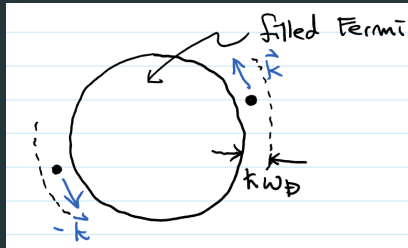


Figure 4: Graphical representation of the Cooper Problem.